

ISSUES OF SYMMETRIES IN THE GROSS-NEVEU MODEL

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We study the chiral Gross-Neveu model using a path integral approach. After showing how $U(1)$ massless excitations decouple we investigate the $SU(N)$ sector of the theory. One then finds that for certain values of the coupling constant, the original (vector $SU(N)$) symmetry is enlarged to a chiral $SU(N)$, giving rise to a conformally invariant model.

The chiral Gross-Neveu model¹ (CGN) is one of the most interesting 2-dimensional fermionic field theories. It shows asymptotic freedom and dynamical mass generation within the $1/N$ approximation¹ without the existence of a Goldstone boson.² The classical Lagrangian has a chiral symmetry which remains unbroken at the quantum level and there is a sector of massless excitations that decouple from the rest of the spectrum.^{3–5} Since it is equivalent to a certain $U(N)$ Thirring model, one can conclude, following Ref. 6 that it also has the peculiarity of being conformally invariant at a special value of the coupling constant.

These properties have been discovered using rather different approaches^{1–11}: $1/N$ expansion, S -matrix construction, Bethe ansatz, current algebra and Sugawara construction, and non-Abelian bosonization. Although they may seem somehow contradictory, all these features rely on the issues of symmetries (particularly chiral symmetry) at the quantum level. It is then useful to examine the CGN model using a path-integral approach^{5,11} which has shown to be very adequate in disclosing the role of symmetries in fermionic models.

It is our purpose in this work to present a discussion on how symmetries are realized in the CGN model using the abovementioned path-integral approach. We first show in a very simple way how massless excitations coming from the $U(1)$ part decouple from the rest and so an “almost long-range order”² à la Kosterlitz-Thouless takes place in the infrared region. We then investigate the $SU(N)$ sector by performing a decoupling change of variables which yields to a bosonic Lagrangian of the Wess-Zumino-Witten⁸ type plus an interaction term. It is then easy to see that for certain values of the coupling constant the original vector $SU(N)$ symmetry is enlarged to chiral $SU(N)$ leading to a conformally invariant model, as first discovered in Ref. 6. In fact, the model becomes at these points a Wess-Zumino-Witten theory at the infrared stable fixed point, giving rise to a Kac-Moody algebra and an associated Virasoro algebra.

We start from the Lagrangian (in 2-dimensional Minkowski space):

$$\mathcal{L} = \bar{\Psi} i \not{\partial} \Psi - \frac{1}{2} g_s^2 j^\mu j_\mu - \frac{1}{2} g_N^2 j^{\mu a} j_\mu^a \quad (1)$$

where ψ^i are N Dirac fermions and j^μ and $j^{\mu a}$ are $U(1)$ and $SU(N)$ currents,

$$\begin{aligned} j^\mu &= \bar{\psi}^i \gamma^\mu \psi^i & i &= 1, 2, \dots, N \\ j^{\mu a} &= \bar{\psi}^i \gamma^\mu t_{ij}^a \psi^j & a &= 1, 2, \dots, N^2 - 1 \end{aligned} \quad (2)$$

with t^a the $SU(N)$ generators,

$$[t^a, t^b] = i f_c^{ab} t^c \quad (3)$$

normalized so that $\text{tr } t^a t^b = \delta^{ab}/2$ and the Casimir operator in the adjoint representation is given by $C(G) = N$.

The special case $g_s^2 = g_N^2/2N$ corresponds to the chiral Gross-Neveu model¹ as can be easily seen by a Fierz transformation. In this case, the model is invariant under global $U(1) \times U(1)$ transformations and also under global $SU(N)$ rotations.

If one is interested in exploring global aspects of the model (such as dynamical mass generation, asymptotic freedom) it is convenient to eliminate quartic interactions by introducing auxiliary scalar fields.¹ Concerning those properties where symmetries play a crucial role, it is preferable to introduce auxiliary vector fields, as explained in Refs. 5 and 11. In particular the g_N^2 term in (1) can be re-written as:

$$\exp \left\{ -i g_{N/2}^2 \int d^2x j^{\mu a} j_\mu^a \right\} = \int DA_\mu \exp \left\{ i \int d^2x \left(\frac{1}{2} A_\mu^a A^{\mu a} - g_N A_\mu^a j^{\mu a} \right) \right\} \quad (4)$$

where $A^\mu = A^{\mu a} t^a$ is a vector field taking values in the Lie algebra of $SU(N)$. A similar identity is used for the $U(1)$ part by introducing an Abelian vector field B^μ . The generating functional for the model then becomes:

$$Z = \int D\bar{\psi} D\psi DA_\mu DB_\mu \exp \left(i \int d^2x (\mathcal{L}_{\text{eff}} + \mathcal{L}_{\text{SOURCES}}) \right) \quad (5)$$

with

$$\mathcal{L}_{\text{eff}} = \bar{\Psi} (i \not{\partial} - g_\mu \not{A} - g_s \not{B}) \Psi + \frac{1}{2} B_\mu B^\mu + \frac{1}{2} A_\mu A^\mu. \quad (6)$$

As explained in Ref. 5 it is very simple now to extract the $U(1)$ part from the fermionic variables by performing the following transformation:

$$\Psi = \exp(i\eta(x) + i\gamma_5\phi(x))\chi(x) = U_0(x)\chi(x) \quad (7a)$$

$$\bar{\Psi} = \bar{\chi}(x) \exp(-i\eta(x) + i\gamma_5\phi(x)) = \bar{\chi}(x) \bar{U}_0(x) \quad (7b)$$

$$B_\mu = -\frac{1}{g_s} (\epsilon_{\mu\nu} \partial^\nu \phi(x) - \partial_\mu \eta(x)). \quad (7c)$$

This transformation completely decouples B_μ from the fermion fields in $\mathcal{L}_{\text{eff}}$. Taken as a change of variables in (5) one has to consider the Fujikawa Jacobian $J_F^{V(1)}$ associated to (7) (the transformation (7c) gives a trivial contribution). One can easily evaluate J_F :^{5,11}

$$J_F = \exp\left(\frac{i}{2\pi} \int d^2x [\partial_\mu \phi \partial^\mu \phi]\right) \exp\left(i \frac{a}{2\pi} \int d^2x B_\mu B^\mu\right) \quad (8)$$

with a an undetermined parameter related to regularization ambiguities.¹² (It is on gauge-invariance grounds that one usually imposes $a = 0$).

It is very simple now to see that massless excitations coming from the U(1) charge decouple from the rest, as predicted in Ref. 2. Indeed, by evaluating for example the 2-point correlation function from Z once the change (7) is performed, one discovers the U(1) factorization:

$$\begin{aligned} \langle \psi(x) \bar{\psi}(0) \rangle &= \langle U_0(x) \chi(x) \bar{\chi}(0) \bar{U}_0(x) \rangle \\ &= \langle U_0(x) U_0^*(0) \rangle_{U(1)} \langle \chi(x) \bar{\chi}(0) \rangle_{\text{SU}(N)} \end{aligned} \quad (9)$$

with the first factor given by:

$$\langle U_0(x) U_0^*(0) \rangle_{U(1)} = \frac{\int D\phi D\eta \exp\left(i \int d^2x \mathcal{L}_{U(1)} U_0(x) U_0^*(0)\right)}{\int D\phi D\eta \exp\left(i \int \mathcal{L}_{U(1)} d^2x\right)} \quad (10)$$

with

$$\mathcal{L}_{U(1)} = \frac{1}{2\pi} \left(1 + a + \frac{\pi}{g_s^2}\right) \partial_\mu \phi \partial^\mu \phi + \frac{1}{2\pi} \left(a + \frac{\pi}{g_s^2}\right) \partial_\mu \eta \partial^\mu \eta. \quad (11)$$

One can exactly evaluate (10):

$$\langle U_0(x) U_0^*(0) \rangle_{U(1)} = (\mu|x|)^{-k(g_s)} \quad (12)$$

with

$$k(g_s) = \frac{g_s^2}{2\pi} \left(1 - \frac{1}{1 + \frac{g_s^2/\pi}{1 + g_s^2 a/\pi}}\right) \frac{1}{1 + g_s^2 a/\pi} \geq 0. \quad (13)$$

One then infers from (9) – (13) that an “almost long range order” occurs² in the infrared region (dominated by massless particles). Note that in an $1/N$ expansion one makes $g_s^2 \rightarrow g_s^2/N$ and hence one recovers through (12) Witten conjecture² on the behavior of the

massless boson propagator for large N . It is interesting to note that this property holds irrespectively of the value of a (provided $a \notin (-1 - \pi/g_s^2, -\pi/g_s^2)$) in accord with the existence of a one parameter family of quantum theories arising from a $U(1)$ Lagrangian of the Thirring type.¹³ Also note that in the ultraviolet, due to asymptotic freedom, $k \rightarrow 0$ and the chirality carrying term goes to one as it should do.

We now turn to the decoupling in the $SU(N)$ sector. We also perform a change of variables as in the $U(1)$ case, now in the form:

$$\begin{aligned}\chi_L &= g^{-1} \eta_L & \chi_R &= h^{-1} \eta_R \\ \bar{\chi}_L &= \bar{\eta}_L g & \bar{\chi}_R &= \bar{\eta}_R h\end{aligned}\quad (14)$$

for the left- and right-handed fermionic components, with g and h elements of $SU(N)$ related to A_μ through:

$$\begin{aligned}A_+ &= -\left(\frac{i}{g_N}\right) g^{-1} \partial_+ g \\ A_- &= -\left(\frac{i}{g_N}\right) h^{-1} \partial_- h.\end{aligned}\quad (15)$$

We then have for the $SU(N)$ factor of the generating functional:

$$\begin{aligned}Z_{SU(N)} &= \int Dg Dh D\bar{\eta} D\eta J(g) J(h) J_F^{SU(N)} \\ &\times \exp\left(-\frac{1}{g_N^2} \text{tr} \int g^{-1} \partial_+ g h^{-1} \partial_- h d^2x\right) \exp\left(i \int \bar{\eta} \not{\partial} \eta d^2x\right)\end{aligned}\quad (16)$$

where:

$$J_F^{SU(N)} = \frac{\det(i\not{\partial} - g_N A)}{\det i\not{\partial}}\quad (17)$$

and:

$$\begin{aligned}DA_+ &= J(g) Dg \\ DA_- &= J(h) Dh.\end{aligned}\quad (18)$$

The fermionic Jacobian can be evaluated using the techniques described in Refs. 5 or 7. The result, first given in Ref. 7 in this form, is:

$$J_F^{SU(N)} = \exp\left\{-iW[g] - i\alpha \text{tr} \int g^{-1} \partial_+ g h^{-1} \partial_- h d^2x\right\}\quad (19)$$

where $W[g]$ is the Wess-Zumino-Witten action:

$$W[g] = \frac{1}{8\pi} \text{tr} \int d^2x \partial_N g^{-1} \partial_N g + \frac{1}{12\pi} \text{tr} \int_B \varepsilon^{ijk} g^{-1} \partial_i g g^{-1} \partial_j g g^{-1} \partial_k g d^3y \quad (20)$$

with the second term integrated over a ball whose boundary is S^2 .⁸

Again, the arbitrary parameter in (19) reflects regularization ambiguities in evaluating $J_F^{SU(N)}$.^{7,11,14-16} Concerning (18), it is easy to see from definition (15) that:

$$J(g) = \det D^{\text{ADJ.}}[A_+] \quad (21)$$

where $D^{\text{ADJ.}}[A_+]$ is the covariant derivative in the adjoint representation. Now, the following relation between determinants for the adjoint and fundamental representation holds:^{7,17-19}

$$\det D^{\text{ADJ.}}[A_+] = (\det D^{\text{FUND.}}[A_+])^{2C(G)} \quad (22)$$

We can then use this result in (19), with $C(G) = N$ both for $J[g]$ and $J[h]$. Collecting all these results we finally have:

$$\begin{aligned} Z_{SU(N)} &= N \int Dg Dh \exp(-i(1 + 2N)W[gh^{-1}]) \\ &\times \exp\left(-i\left[\frac{1}{g_N^2} + \alpha(1 + 2N)\right] \text{tr} \int d^2x g^{-1} \partial_+ g h^{-1} \partial_- h\right) \\ &\times \exp\left(i(2N + 1) \int d^2x \bar{\eta} \not{\partial} \eta D\bar{\eta} D\eta\right). \end{aligned} \quad (23)$$

We shall now show how this generating functional becomes that of a conformal invariant theory for certain critical values of the coupling constant. Indeed, the second exponential in (23) can be adjusted so as to vanish,

$$\frac{1}{g_N^2} + (1 + 2N)\alpha = 0 \quad (24)$$

then, by making $gh^{-1} = U$ one gets:

$$Z_{SU(N)} = N' \int DU e^{-i(1+2N)W[U]} \quad (25)$$

since the h integration becomes trivial. Now, (25) shows an invariance with respect to an infinite dimensional Kac-Moody algebra⁸ since $W[U]$ is invariant under the transformation

$$U(x) \rightarrow \Omega(X^+) U(x) \Lambda(X^-) \quad (26)$$

i.e. local $SU(N) \times SU(N)$ transformations (with Ω and Λ depending on X^+ and X^- respectively). We then see that for any choice of the regularization parameter α there exists a particular value of the coupling constant g_N , which we call g_N^* ,

$$g_N^{*2} = -\frac{1}{\alpha(1 + 2N)} \quad (27)$$

for which the CGN model has a local $SU(N) \times SU(N)$ symmetry giving rise to an infinite number of conserved currents which can be evaluated from the basic ones:

$$J_+ = \frac{K}{i} U^{-1} \partial_+ U \quad (28)$$

$$J_- = \frac{K}{i} U \partial_- U^{-1} \quad K = 2N + 1$$

obeying the conservation equations:

$$\partial_+ J_- = \partial_- J_+ = 0. \quad (29)$$

These currents then satisfy a Kac-Moody algebra with level $k = 2N + 1$. Moreover, since $W[U]$ corresponds to a Wess-Zumino-Witten action at the infrared-stable point $\lambda^2 = 4\pi/K$,⁸ we conclude that at $g_N = g_N^*$ the model exhibits conformal invariance, the energy-momentum tensor components satisfying a Virasoro algebra with central charge $c = (k \dim G)/[k + C(G)]$. Taking into account the $U(1)$ and free fermionic contribution to c , we have

$$c = \frac{(2N + 1)(N^2 - 1)}{(3N + 1)} + 1 + N(2N + 1).$$

The low-energy excitations of the theory are then fixed by these algebras and the conservation equations obeyed by $J_\pm$ as explained in Ref. 10.

If instead of writing Z_N as in (23) we use the well-known identity⁷

$$W[gh^{-1}] = W[g] + W[h^{-1}] + \frac{1}{4\pi} \int d^2x \operatorname{tr}(g^{-1} \partial_+ g)(h^{-1} \partial_- h) \quad (30)$$

we then have:

$$\begin{aligned} Z_{SU(N)} = N \int Dg Dh \exp(-i(2N + 1)[W[g] + W[h^{-1}]]) \\ \times \exp\left(-i\left[\frac{1}{g_N^2} + (2N + 1)\left(\frac{1}{4\pi} + \alpha\right)\right] \int \operatorname{tr}(g^{-1} \partial_+ g)(h^{-1} \partial_- h) d^2x\right). \end{aligned} \quad (31)$$

Hence, for $g_N = g_N^{**}$ satisfying:

$$\frac{1}{g_N^{**2}} + (2N + 1)(1/4\pi + \alpha) = 0 \quad (32)$$

$Z_{SU(N)}$ takes the form:

$$Z_{SU(N)} = N \int e^{-i(2N+1)W[g]} Dg \int e^{-i(2N+1)W[h^{-1}]} Dh. \quad (33)$$

Then, for $g_N = g_N^{**}$ $Z_{SU(N)}$ becomes the generating functional for two Wess-Zumino-Witten models at the infrared stable point. Note that in this case, $Z_{SU(N)}$ has the symmetry $g \rightarrow g\Omega$, $h \rightarrow h\Lambda$ which is precisely related to chiral transformations of the original fermi fields, $\psi_L \rightarrow \Omega^{-1}\psi_L$, $\psi_R \rightarrow \Lambda^{-1}\psi_R$ respectively. (This can easily be seen by analysing the transformation properties of j_μ and A_μ in identity (4)).

We then conclude that although the $SU(N)$ current-current interaction breaks the chiral $SU(N) \times SU(N)$ of free fermions in the chiral Gross-Neveu model, there are two critical values of $g_N \neq 0$ for which conformal invariance reappears in the interacting model which exhibits a larger (chiral) symmetry.

This phenomenon was first observed by Dashen and Frishman.⁶ In fact, if we choose the regularization which ensures the equivalence between free fermions and Wess-Zumino-Witten theories,¹⁴⁻¹⁶ then $\alpha = -1/4\pi$ and g_N^* becomes the critical point discovered by Dashen and Frishman in Ref. 6:

$$g_N^{*2} = \frac{4\pi}{2N + 1} \quad (34)$$

(In Ref. 6 there is a $(N + 1)$ in the denominator of Eq. (34)).

Note that for this regularization, there is no finite g_N^{**} satisfying (32). Notice also that at the critical points, the theory is not a free one since $k = 2N + 1 \neq 1$ for any value of α .

Of course, dynamical mass generation cannot take place at $g_N = g_N^*$, g_N^{**} ; this is discussed in Ref. 20 where the existence of a zero in the β function is studied.

Outside the critical points, one can re-write (23) in the form:

$$Z_{SU(N)} = \int DU Dh e^{-i(2N+1)W[U]} \times \exp\left(-i\left(\frac{1}{g_N^2} - \frac{1}{g_N^{*2}}\right) \text{tr} \int d^2x (h^{-1}\partial_+ h h^{-1}\partial_- h + U^{-1}\partial_+ U h^{-1}\partial_- h)\right) \quad (30)$$

and evaluate the Poisson brackets for $J_\pm$ following Ref. 8. One then obtains a Kac-Moody like algebra for J_+ but it is only for $g_N = g_N^*$ that currents are conserved and conformal invariance holds.

We end this work by noting that recently²¹ it has been shown that in two-dimensional field theories there exists a function $c(g)$ related to the β function which at the fixed points (for which $\beta = 0$) coincides with the central charge of the Virasoro algebra of the corresponding conformal field theory. We think the behavior of the CGN model we have described provides a concrete example of this phenomenon and we hope to return to this point in a future work.

Note added

After this work was completed, we become aware of Refs. 22–23 where similar aspects of the CGN model are discussed. In Ref. 22, expressions for the critical points like those in Eqs. (32), and (34) were derived. In Ref. 23 the same authors criticized the derivation of the Jacobian $J[g]$, leading to these critical points. The reasons why $J[g]$ is given by Eq. (21), exposed in the present work, are also explained in Refs. 7, 17, 18, 19 and 22. The choice in Ref. 23 ($J[g] = 1$) is not consistent with any regularization for fermion determinants leading to the Wess-Zumino-Witten action.

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